

Assignment 5 Implementation Report

Technical Analysis using R Development Phase

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Course	BDA400 Data Science Tools and Techniques
Repository	https://github.com/samyar11293/TechnicalAnalysis/tree/main/Assignment05

Implementation approach

The submission implements every required indicator directly with standard base-R operations. It does not call technical-analysis packages or prebuilt indicator functions. Each function is isolated in its own script, uses the exact required function and argument names, validates invalid inputs, and returns the structure described in the assignment templates.

How to run the project

- Download the Assignment05 folder and open it in RStudio.
- Set the working directory to the folder containing the R scripts.
- Run `source("run_tests.R")` to load every function and execute all checks.
- For MACD, source `ema.R` first. For Stochastic RSI, source `sma.R` and `rsi.R` first.
- Use `source("filename.R")` to load any individual implementation.

SMA Simple Moving Average

Required function: `sma(data, period)`

SMA smooths a numeric series by assigning equal weight to every observation in each rolling window.

Implementation	A loop extracts every window of period observations and divides its sum by period.
Returned result	A numeric vector of length $n - \text{period} + 1$.
Validation	Rejects nonnumeric or empty data, invalid periods, and periods longer than the data.
File	<code>sma.R</code>

EMA Exponential Moving Average

Required function: `ema(data, period)`

EMA emphasizes recent observations with multiplier $2 / (\text{period} + 1)$.

Implementation	The first EMA equals the first observation; subsequent values apply the recursive assignment formula from the pseudocode.
Returned result	A numeric vector with the same length as data.
Validation	Rejects nonnumeric or empty data and nonpositive or noninteger periods.
File	ema.R

MACD Moving Average Convergence Divergence

Required function: `macd(data, short_period, long_period, signal_period)`

MACD measures trend momentum from the difference between short and long EMAs.

Implementation	The function calculates both EMAs, subtracts them for the MACD line, applies EMA to create the signal line, and subtracts again for the histogram.
Returned result	A named list containing <code>macd_line</code> , <code>signal_line</code> , and <code>histogram</code> .
Validation	Requires <code>ema()</code> , positive integer periods, and a short period smaller than the long period.
File	macd.R

STDEV Population Standard Deviation

Required function: `stdev(data)`

Standard deviation measures dispersion around the arithmetic mean using the population denominator n required by the assignment.

Implementation	The function calculates the mean, deviations, squared deviations, population variance, and its square root.
Returned result	One nonnegative numeric value.
Validation	Rejects nonnumeric or empty input.
File	stdev.R

LINREG Linear Regression

Required function: `linreg(regressionSource, regressionLength, regressionOffset)`

Linear regression estimates the slope and intercept of the selected sequence window.

Implementation	The implementation selects the required subset, derives index values, and applies the ordinary least-squares numerator and denominator from the pseudocode.
Returned result	A named list containing slope, intercept, and predicted_values.
Validation	Checks source type, length, offset range, integer arguments, and degenerate windows.
File	linreg.R

RSI Relative Strength Index

Required function: `rsi(data, period)`

RSI expresses recent upward and downward price movement on a 0-to-100 scale.

Implementation	Price differences are separated into gains and losses. Initial period averages and Wilder smoothing produce the RSI series; zero-gain and zero-loss cases are handled explicitly.
Returned result	A vector aligned to the input, with NA before the first calculable RSI.
Validation	Checks numeric data, positive integer period, sufficient history, and stable division edge cases.
File	rsi.R

Stochastic RSI

Required function: `stoch_rsi(data, period, k_period, d_period)`

StochRSI normalizes valid RSI values and smooths them into percent K and percent D lines.

Implementation	The valid RSI range is normalized to 0 through 1, SMA generates the K line, and another SMA generates the D line.
Returned result	A named list containing <code>k_line</code> and <code>d_line</code> .
Validation	Requires <code>rsi()</code> and <code>sma()</code> , validates periods and sample sufficiency, and handles a zero RSI range.
File	<code>stoch_rsi.R</code>

Crossover Signals

Required function: `crossover(arr1, arr2)`

Crossover identifies upward and downward intersections between equal-length numeric arrays.

Implementation	Every position after the first compares the current and previous relationship and returns Up, Down, or None.
Returned result	A character vector with the same length as the inputs.
Validation	Checks equal lengths and nonempty numeric vectors.
File	<code>crossover.R</code>

Crossunder Signals

Required function: `crossunder(arr1, arr2)`

Crossunder identifies positions where the first series moves from at or above the second series to below it.

Implementation	Every position after the first checks the specified current and previous comparison.
Returned result	A character vector containing None for the first value and True or False thereafter.
Validation	Checks equal lengths and nonempty numeric vectors.
File	<code>crossunder.R</code>

Verification strategy

The included `run_tests.R` script sources all nine files and checks exact SMA values, EMA and MACD lengths, MACD list structure, population standard deviation, regression coefficients, RSI and StochRSI ranges, and known crossover and crossunder events. Any failed assertion stops execution and names the failed test; successful execution ends with All Assignment 5 verification tests passed.

Assumptions

- Input series are numeric, ordered from oldest to newest, and contain no missing values unless a function creates leading NA values by design.
- `stdev()` uses the population formula with denominator n , matching the assignment formula rather than R's sample `sd()` convention.
- SMA output is shortened to the number of complete windows, matching the supplied pseudocode.
- RSI returns leading NA values because RSI is undefined until enough price changes exist.
- These educational indicators demonstrate algorithmic implementation and are not financial advice.